

# Assignment 2 Complete Solutions

## Discrete Random Variables

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### Problem 1

**Solution.** (a)

$$P(F) = 0.4 \sum_{k=0}^9 (0.3)^k = 0.4 \frac{1 - 0.3^{10}}{0.7}$$

(b)

$$P(T = k) = 0.7(0.3)^{k-1}, \quad k = 1, \dots, 9$$

$$P(T = 10) = 0.7(0.3)^9 + (0.3)^{10}$$

| $k$ | $P(T = k)$                |
|-----|---------------------------|
| 1   | 0.7                       |
| 2   | 0.21                      |
| 3   | 0.063                     |
| 4   | 0.0189                    |
| 5   | 0.00567                   |
| 6   | 0.001701                  |
| 7   | 0.0005103                 |
| 8   | 0.00015309                |
| 9   | 0.000045927               |
| 10  | $0.7(0.3)^9 + (0.3)^{10}$ |

□

### Problem 2

**Solution.** (a)

$$P(X = 1) = \binom{500}{1} \frac{1}{365} \left( \frac{364}{365} \right)^{499}$$

(b)

$$\lambda = \frac{500}{365}, \quad P(X = 1) \approx e^{-\lambda} \lambda$$

□

### Problem 3

**Solution.** (a)

$$P(X = k) = \binom{1000}{k} (0.01)^k (0.99)^{1000-k}$$

(b)

$$\lambda = 10, \quad P(X = k) \approx e^{-10} \frac{10^k}{k!}$$

(c)

$$\begin{aligned} P(X > 50) &= \sum_{k=51}^{1000} \binom{1000}{k} (0.01)^k (0.99)^{1000-k} \\ &\approx \sum_{k=51}^{\infty} e^{-10} \frac{10^k}{k!} \\ P(X > 50) &\approx 0 \end{aligned}$$

**Remark:** Since the mean is 10, the probability of exceeding 50 is extremely small (practically zero).  $\square$

### Problem 4

**Solution.**

(a) Without replacement:

$$P(T = k) = \frac{1}{5}, k = 1, \dots, 5$$

With replacement:

$$P(T = k) = \left(\frac{4}{5}\right)^{k-1} \frac{1}{5}$$

(b)

$$P(T = k) = \left(\frac{2(10-x)}{90}\right)$$

$\square$

### Problem 5

**Solution.**

All possible values of  $(X, Y)$  satisfy  $x + y \leq 4$ .

$$P(X = x, Y = y) = \frac{4!}{x!y!(4-x-y)!} \left(\frac{1}{6}\right)^{x+y} \left(\frac{2}{3}\right)^{4-x-y}$$

| $(x, y)$ | $P(X = x, Y = y)$                                                |
|----------|------------------------------------------------------------------|
| $(0, 0)$ | $\left(\frac{2}{3}\right)^4$                                     |
| $(1, 0)$ | $4 \cdot \frac{1}{6} \left(\frac{2}{3}\right)^3$                 |
| $(0, 1)$ | $4 \cdot \frac{1}{6} \left(\frac{2}{3}\right)^3$                 |
| $(1, 1)$ | $12 \cdot \left(\frac{1}{6}\right)^2 \left(\frac{2}{3}\right)^2$ |
| $(2, 0)$ | $6 \cdot \left(\frac{1}{6}\right)^2 \left(\frac{2}{3}\right)^2$  |
| $(0, 2)$ | $6 \cdot \left(\frac{1}{6}\right)^2 \left(\frac{2}{3}\right)^2$  |
| $(2, 1)$ | $12 \cdot \left(\frac{1}{6}\right)^3 \left(\frac{2}{3}\right)$   |
| $(1, 2)$ | $12 \cdot \left(\frac{1}{6}\right)^3 \left(\frac{2}{3}\right)$   |
| $(3, 0)$ | $4 \cdot \left(\frac{1}{6}\right)^3 \left(\frac{2}{3}\right)$    |
| $(0, 3)$ | $4 \cdot \left(\frac{1}{6}\right)^3 \left(\frac{2}{3}\right)$    |
| $(4, 0)$ | $\left(\frac{1}{6}\right)^4$                                     |
| $(0, 4)$ | $\left(\frac{1}{6}\right)^4$                                     |

□

## Problem 6

**Solution.** (a)

$$a = \sum_{x=-3}^3 x^2 = 28$$

Symmetry gives  $E[X] = 0$ .

(b) Values of  $Z = X^2$  give PMF accordingly.

(c)

$$\text{Var}(X) = E[X^2] = \frac{\sum x^4}{28} = \frac{196}{28} = 7$$

(d) Same result.

□

## Problem 7

**Solution.**

Let  $X$  be uniformly distributed on  $\{0, 1, 2, \dots, 9\}$ , i.e.,

$$P(X = x) = \frac{1}{10}, \quad x = 0, 1, \dots, 9.$$

**(a) PMF of  $Y = X \bmod 3$ :**

Compute values:

$$X \bmod 3 = 0 : X = 0, 3, 6, 9 \quad (4 \text{ values})$$

$$X \bmod 3 = 1 : X = 1, 4, 7 \quad (3 \text{ values})$$

$$X \bmod 3 = 2 : X = 2, 5, 8 \quad (3 \text{ values})$$

Thus,

$$P(Y = 0) = \frac{4}{10}, \quad P(Y = 1) = \frac{3}{10}, \quad P(Y = 2) = \frac{3}{10}.$$

**(b) PMF of  $Y = 5 \bmod (X + 1)$ :**

Compute for each  $x$ :

| $X$ | $Y = 5 \bmod (X + 1)$ |
|-----|-----------------------|
| 0   | $5 \bmod 1 = 0$       |
| 1   | $5 \bmod 2 = 1$       |
| 2   | $5 \bmod 3 = 2$       |
| 3   | $5 \bmod 4 = 1$       |
| 4   | $5 \bmod 5 = 0$       |
| 5   | $5 \bmod 6 = 5$       |
| 6   | $5 \bmod 7 = 5$       |
| 7   | $5 \bmod 8 = 5$       |
| 8   | $5 \bmod 9 = 5$       |
| 9   | $5 \bmod 10 = 5$      |

Count occurrences:

$Y = 0$  : 2 times

$Y = 1$  : 2 times

$Y = 2$  : 1 time

$Y = 5$  : 5 times

Thus,

$$P(Y = 0) = \frac{2}{10}, \quad P(Y = 1) = \frac{2}{10}, \quad P(Y = 2) = \frac{1}{10}, \quad P(Y = 5) = \frac{5}{10}.$$

□

## Problem 8

**Solution.**

The joint PMF of  $(X, Y)$  is uniform over the set

$$S = \{(x, y) : -2 \leq x \leq 4, -1 \leq y - x \leq 1\}.$$

For each integer  $x \in \{-2, -1, 0, 1, 2, 3, 4\}$ , the condition

$$-1 \leq y - x \leq 1 \Rightarrow y \in \{x - 1, x, x + 1\}$$

gives 3 possible values of  $y$ . Hence, total outcomes:

$$|S| = 7 \times 3 = 21.$$

Thus,

$$p_{X,Y}(x,y) = \frac{1}{21}.$$

**(a) Marginal PMFs and means**

**Marginal of  $X$ :**

$$p_X(x) = \sum_y p_{X,Y}(x,y) = 3 \cdot \frac{1}{21} = \frac{1}{7}, \quad x = -2, -1, 0, 1, 2, 3, 4.$$

Thus  $X$  is uniform on  $\{-2, \dots, 4\}$  and

$$\mathbb{E}[X] = \frac{-2 + 4}{2} = 1.$$

**Marginal of  $Y$ :**

List possible values of  $Y$ :

| $y$ | Number of pairs |
|-----|-----------------|
| -3  | 1               |
| -2  | 2               |
| -1  | 3               |
| 0   | 3               |
| 1   | 3               |
| 2   | 3               |
| 3   | 3               |
| 4   | 2               |
| 5   | 1               |

Thus,

$$p_Y(y) = \frac{\text{count}}{21}.$$

Compute expectation:

$$\mathbb{E}[Y] = \frac{1}{21}(-3 - 4 - 3 + 0 + 3 + 6 + 9 + 8 + 5) = 1.$$

|                                              |
|----------------------------------------------|
| $\mathbb{E}[X] = 1, \quad \mathbb{E}[Y] = 1$ |
|----------------------------------------------|

**(b) Mean total profit**

Total profit:

$$Z = 100X + 200Y.$$

Using linearity of expectation:

$$\mathbb{E}[Z] = 100\mathbb{E}[X] + 200\mathbb{E}[Y] = 100(1) + 200(1) = 300.$$

|                       |
|-----------------------|
| $\mathbb{E}[Z] = 300$ |
|-----------------------|

□

## Problem 9

**Solution.** (a)

$$E[X] = np = 5$$

$$P(X = 5) = \binom{250}{5} (0.02)^5 (0.98)^{245}$$

(b)

$$\lambda = 5, P(X = 5) \approx e^{-5} \frac{5^5}{5!}$$

(c)

$$E[Ticket] = 0.5(10) + 0.3(20) + 0.2(50) = 19$$

$$Var = E[X]Var(T) + Var(X)(E[T])^2$$

(d)

$$\hat{p} = \frac{5}{250} = 0.02$$

□

## Problem 10

**Solution.** Sunday:

$$\lambda = 130 \times 240$$

$$P = e^{-\lambda}$$

Bayes theorem for second part.

□

## Problem 11

**Solution.**

(a) We use the classical combinatorial probability formula:

$$P = \frac{|A|}{|S|}$$

where  $A$  is the set of favorable outcomes and  $S$  is the sample space.

Let  $X_r$  be the number of red balls drawn when 10 balls are selected from 50 (with 20 red and 30 non-red). Then:

$$P(X_r = 4) = \frac{\binom{20}{4} \binom{30}{6}}{\binom{50}{10}} \approx 0.28.$$

(b) We want to compute:

$$P(X_r = 4 \mid X_r \geq 3).$$

Using conditional probability:

$$P(X_r = 4 \mid X_r \geq 3) = \frac{P(X_r \geq 3 \mid X_r = 4) P(X_r = 4)}{P(X_r \geq 3)}.$$

Since  $X_r = 4 \Rightarrow X_r \geq 3$ , we have:

$$P(X_r \geq 3 \mid X_r = 4) = 1,$$

so:

$$P(X_r = 4 \mid X_r \geq 3) = \frac{P(X_r = 4)}{P(X_r \geq 3)}.$$

Now,

$$P(X_r \geq 3) = 1 - P(X_r = 0) - P(X_r = 1) - P(X_r = 2).$$

Thus:

$$\begin{aligned} P(X_r = 4 \mid X_r \geq 3) &= \frac{\binom{20}{4} \binom{30}{6}}{\binom{50}{10} - \binom{20}{0} \binom{30}{10} - \binom{20}{1} \binom{30}{9} - \binom{20}{2} \binom{30}{8}} \\ &\approx 0.33. \end{aligned}$$

□

## Problem 12

**Solution.**

(a) Let  $X'$  be the number of tosses until the game ends. Then  $X'$  follows a geometric distribution with parameter  $p = \frac{1}{2}$ :

$$P(X' = k) = \left(\frac{1}{2}\right)^k, \quad k = 1, 2, 3, \dots$$

Let  $X$  be the winnings, where

$$X = 2^{k-1}, \quad k = 1, 2, 3, \dots$$

Thus,

$$P(X = 2^{k-1}) = P(X' = k) = \left(\frac{1}{2}\right)^k$$

The expected value is:

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} 2^{k-1} \left(\frac{1}{2}\right)^k = \sum_{k=1}^{\infty} \frac{1}{2} = \infty$$

$$\boxed{\mathbb{E}[X] = \infty}$$

(b) We need:

$$P(X > 65)$$

Since

$$2^6 = 64 < 65, \quad 2^7 = 128 > 65$$

the condition is  $k \geq 8$ . Hence:

$$P(X > 65) = P(X' \geq 8) = \sum_{k=8}^{\infty} \left(\frac{1}{2}\right)^k$$

Using geometric series:

$$= \frac{(1/2)^8}{1 - 1/2} = \frac{1}{128}$$

$$\boxed{P(X > 65) = \frac{1}{128}}$$

(c) Now suppose the payout is capped at  $2^{30}$ . Define:

$$Y = \begin{cases} 2^{k-1}, & k \leq 31, \\ 2^{30}, & k \geq 32 \end{cases}$$

Then:

$$\mathbb{E}[Y] = \sum_{k=1}^{31} 2^{k-1} \left(\frac{1}{2}\right)^k + \sum_{k=32}^{\infty} 2^{30} \left(\frac{1}{2}\right)^k$$

First term:

$$\sum_{k=1}^{31} \frac{1}{2} = \frac{31}{2}$$

Second term:

$$\sum_{k=32}^{\infty} \left(\frac{1}{2}\right)^k = \frac{(1/2)^{32}}{1 - 1/2} = \frac{1}{2^{31}}$$

Thus:

$$2^{30} \cdot \frac{1}{2^{31}} = \frac{1}{2}$$

$$\mathbb{E}[Y] = \frac{31}{2} + \frac{1}{2} = 16$$

$$\boxed{\mathbb{E}[Y] = 16}$$

□